

# Causal ML, Policy Learning, and External Validity

## Empirical Methods — Week 11

Teaching deck draft

## Week 11 question

- What should researchers do after heterogeneity is estimated?
- When is heterogeneity itself the research object?
- When is the object a policy rule, value, or regret comparison?
- When does a rule learned in one setting fail to travel to another?

The method is useful only after the welfare objective, feasible rule class, and deployment environment are explicit.

# Lecture 10 versus Lecture 11

| Lecture 10                      | Lecture 11                                      |
|---------------------------------|-------------------------------------------------|
| High-dimensional controls       | Policy rules and targeting                      |
| Orthogonal nuisance estimation  | Policy value and regret                         |
| DML for low-dimensional effects | Empirical welfare maximization                  |
| CATE and causal forests         | CATEs as inputs to decisions                    |
| Overlap for effect estimation   | Overlap where the rule assigns treatment        |
| Inference for effects           | Validation for value, portability, and fairness |

The continuity is real, but the target object changes.

## Three objects that often get confused

| Object        | Mathematical object                            | Applied question                           |
|---------------|------------------------------------------------|--------------------------------------------|
| Prediction    | $\hat{m}(x) \approx \mathbb{E}[Y \mid X = x]$  | What outcome or risk do we expect?         |
| Causal effect | $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ | For whom does treatment change outcomes?   |
| Policy rule   | $\pi(x) \in \{0, 1\}$                          | Who should be assigned, under constraints? |

High predicted outcomes are not the same as high treatment effects.

# CATE versus policy rule

$$\tau_i = Y_i(1) - Y_i(0), \quad \tau_G = \mathbb{E}[Y(1) - Y(0) \mid X \in G]$$

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x], \quad \pi(x) \in \{0, 1\}$$

- CATEs summarize heterogeneity under an identifying design.
- Subgroup effects answer incidence or mechanism questions.
- A policy rule adds welfare weights, costs, capacity, legality, information, and implementation.
- The simple rule  $\pi^*(x) = \mathbf{1}\{\tau(x) \geq 0\}$  needs costless treatment and no constraints.

$$V(\pi) = \mathbb{E}[Y(\pi(X))] = \mathbb{E}[\pi(X)Y(1) + (1 - \pi(X))Y(0)]$$

$$V^{net}(\pi) = \mathbb{E}[\pi(X)(Y(1) - c(X)) + (1 - \pi(X))Y(0)]$$

$$R(\pi) = V(\pi^*) - V(\pi)$$

- Report value relative to treat-all, treat-none, simple rules, and current assignment.
- Evaluate value on held-out or cross-fit scores, not only in the sample used to choose the rule.
- A stable simple rule can have low regret even when a flexible rule has higher in-sample value.

# Empirical welfare maximization

$$\hat{\pi} \in \arg \max_{\pi \in \Pi} \hat{V}(\pi)$$

- $\Pi$  is the feasible policy class: thresholds, trees, score rules, or constrained assignment rules.
- $\hat{V}(\pi)$  estimates welfare or net outcomes under the rule.
- Capacity, budget, and fairness constraints belong in  $\Pi$ , not in an afterthought paragraph.
- Kitagawa–Tetenov and Athey–Wager treat rules as empirical objects whose value can be estimated.

Policy learning asks which rule performs best, not which subgroup has the largest noisy effect estimate.

## Doubly robust policy value

$$\mu_d(x) = \mathbb{E}[Y \mid D = d, X = x], \quad e(x) = \Pr(D = 1 \mid X = x)$$

$$\begin{aligned} \Gamma_i(\pi) &= \pi(X_i) \left[ \hat{\mu}_1(X_i) + \frac{D_i}{\hat{e}(X_i)} (Y_i - \hat{\mu}_1(X_i)) \right] \\ &+ (1 - \pi(X_i)) \left[ \hat{\mu}_0(X_i) + \frac{1 - D_i}{1 - \hat{e}(X_i)} (Y_i - \hat{\mu}_0(X_i)) \right] \\ \hat{V}_{DR}(\pi) &= \frac{1}{n} \sum_i \Gamma_i(\pi) \end{aligned}$$

This inherits Lecture 10's discipline: identify first, learn nuisance functions honestly, and evaluate the target on held-out scores.



# External validity and transportability

$$V_T(\pi) = \mathbb{E}_T[Y(\pi(X))]$$

$$\mathbb{E}_T[h(X)] = \mathbb{E}_S[\omega(X)h(X)], \quad \omega(X) = \frac{f_T(X)}{f_S(X)}$$

$$V_T(\pi) \approx \mathbb{E}_S [\omega(X)\{\mu_0(X) + \pi(X)\tau(X)\}]$$

- Reweighting helps only when relevant state variables are observed and support overlaps.
- Target populations may differ in effects, take-up, costs, implementation, prices, or equilibrium response.
- External validity is a design claim, not a property of an algorithm.

# Fairness and implementation constraints

- Is every feature observable before assignment?
- Is the rule legal and administratively feasible?
- Does assignment create disparate impact across protected groups?
- Are protected groups proxied by geography, language, school, firm, or case history?
- Can workers, firms, schools, or staff game the rule?
- Does targeting change take-up, stigma, search, or local equilibrium behavior?

Statistical optimality does not settle legitimacy, feasibility, or welfare.

# ML algorithm quick guide

| Tool                      | Most suitable for                            | Main applied caveat                              |
|---------------------------|----------------------------------------------|--------------------------------------------------|
| Lasso / elastic net       | high-dimensional controls, sparse prediction | selected variables are not mechanisms            |
| Ridge                     | dense correlated predictors                  | weak interpretability                            |
| Random forests / boosting | nonlinear tabular prediction                 | feature importance is not causal importance      |
| Causal forests / GRF      | heterogeneous effects under a design         | CATEs are not rules by themselves                |
| Policy trees              | interpretable assignment rules               | unstable when samples are small                  |
| SVM                       | margin-based classification                  | not a natural welfare or causal tool             |
| Neural nets               | text, images, sequences, very large data     | opaque and often excessive for tabular economics |
| Clustering                | exploratory typologies                       | clusters are not causal groups                   |

## Reproduce

- synthetic job-training data
- CATE by subgroup
- feasible policy class
- value and regret

## Diagnose

- held-out value
- overlap in targeted group
- capacity sensitivity
- fairness audit

## Transfer

- shifted target population
- source-target weights
- target value comparison
- portability memo

The lab reproduces policy-learning logic, not official paper estimates.

# Takeaways

- Heterogeneity is not automatically a policy.
- Policy learning needs a rule class, value function, costs, constraints, and honest evaluation.
- Targeting should be based on causal gains, not predicted outcome levels.
- External validity gets harder when effects and institutions vary.
- Fairness and implementation are part of applied economics, not decoration after the estimator.
- The best empirical recommendation may be a modest rule with clear value and low regret.